Lie Detector Over Media

Software Requirements Specification (SRS) - 'See what's real.'

Author & Project Lead: Sarvesh Pandey | Copyright (c) 2026 Sarvesh Pandey
License: MIT License | Version: 1.0.0 | Status: Verified & Approved Production Release
Official Repository: https://github.com/xsarveshpandey21/lie-detector-over-media

1. Executive Summary & Core Principles

Lie Detector Over Media is an open, transparent educational verification platform engineered to help everyday users evaluate the credibility and authenticity of digital media across five primary modalities: Text statements, Photos, Videos, Voice/Audio recordings, and Website Links. The product translates complex forensic metadata and computational linguistics into plain, accessible English and Hindi without confusing jargon.

GOVERNING SYSTEM PHILOSOPHY

Never invent or fake data. Never claim 100% certainty. The tool provides explainable signals, not final proof. When an external neural model or third-party API is unconfigured, the system transparently reports 'Not checked' with score: null rather than generating deceptive placeholder numbers or simulated ratings.

- **Core User Interface Flow:** The user experience is strictly split into (1) What you gave us, and (2) What we found.
- **Zero Persistent Storage:** Uploaded media and pasted claims are processed solely in volatile memory and never written to disk.
- **Auditability & Transparency:** Every analysis includes an explicit check ledger specifying each inspection that ran or was skipped.

2. System Architecture & Runtime Stack

The system employs a modular, decoupled full-stack architecture that supports local development and serverless deployment on Vercel with zero runtime dependency conflicts.

Layer	Technology / Framework	Specification & Architectural Role
Frontend	React 18 + Vite + TS	TypeScript strict mode (0 any), Tailwind CSS, warm accessible paper theme
Backend API	Node.js + Express (CJS)	Express 4.19, Multer in-memory (4MB limit), Helmet security, CORS, Rate Limit
ML Microservice	Python 3.10+ & FastAPI	FastAPI, PyTorch, Transformers (DistilBERT / RoBERTa trained on LIAR)
Deployment	Vercel Serverless Function	api/index.js entrypoint, vercel.json rewrites /api/* to Express router
Internationalization	Native i18n JSON	Comprehensive dual-language support for English (en) and Hindi (hi)

Copyright (c) 2026 Sarvesh Pandey. All rights reserved. (MIT License)

3. Functional Requirements Specification

3.1 Media Modality Forensic Checks

- **FR-1.1 Text Credibility:** Evaluates sensationalism, artificial urgency, source attributions, reading ease (Flesch-Kincaid), and lexical diversity. Integrates optional local Hugging Face transformer classification.
- **FR-1.2 Image Forensics:** Performs binary magic-byte inspection (JPEG, PNG, WebP, GIF), parses camera hardware EXIF tags (Make, Model, GPS), and flags image manipulation software tags (Photoshop, Canva, GIMP).
- **FR-1.3 Video Integrity:** Inspects ISO MP4 and WebM container atom hierarchies, validates video stream MIME consistency, and verifies stream metadata without client-side script execution.
- **FR-1.4 Audio Forensics:** Analyzes audio containers, duration, sample rate (44.1/48 kHz), bitrate, and channel configurations via music-metadata to flag synthetic or abnormal compression traces.
- **FR-1.5 Website Link Verification:** Verifies TLS/HTTPS encryption, identifies raw IP addresses, warns on URL shorteners (bit.ly, tinyurl), detects Punycode/IDN homograph attacks, and checks lookalikes against trusted news domains.

3.2 Accurate Claim Matching Engine (False-Positive Elimination)

To prevent harmless news or debunks from being falsely flagged, the system enforces a strict multi-phrase matcher (backend/src/services/claimMatcher.js):

- **FR-2.1 Text Normalization:** Converts text to lowercase, collapses whitespace, and strips punctuation including em/en dashes and Hindi danda (।).
- **FR-2.2 Multi-Word Phrases:** Registry entries must contain multi-word compound phrases (e.g. '5g towers spread', 'apollo moon landing faked') with generic isolated words excluded.
- **FR-2.3 Strict Match Threshold:** Requires matched phrase weight / total weight >= 0.60 AND at least 3 distinct phrase matches (or 2 if total phrases <= 3).
- **FR-2.4 Debunking Rejection:** Automatically rejects matches if input text contains debunking terms (disproves, myth, false claim, debunked, busted, jhootha). False-positive rate on benign claims must be <= 2% (achieved 0.00%).
- **FR-2.5 Honest Attribution:** Registry matches must strictly be presented as 'Similar claim previously fact-checked', never 'this claim'. Only live Google Fact Check API matches can confirm identical claims.

3.3 Confidence Assessment Engine

- **FR-3.1 High Confidence:** Assigned only upon an identical live Google Fact Check API match OR unanimous agreement between semantic similarity (cosine >= 0.78) and NLI entailment.
- **FR-3.2 Medium Confidence:** Assigned for strong registry phrase matches (>= 80%) OR verified hardware EXIF tags OR clear stylistic signals corroborated by an external model.
- **FR-3.3 Low Confidence:** Assigned when only linguistic heuristics exist without external corroboration, or when analyzing basic file container headers alone.

Copyright (c) 2026 Sarvesh Pandey. All rights reserved. (MIT License)

4. Data Schema & Verified References Specification

4.1 Core API Response Structure (/api/analyze)

The backend implements a standardized, self-documenting JSON contract:

Field	Type	Description / Constraints
type	string	Enum: 'text' | 'image' | 'video' | 'audio' | 'url'
verdict.label	string	Enum: 'likely_true' | 'likely_false' | 'uncertain'
verdict.score	number | null	0-100 authenticity score; null if file alone cannot prove validity
verdict.confidenceLevel	string	Enum: 'high' | 'medium' | 'low'
confidence	object	Structured assessment with level, score (0-100), and plain reason
reasons[]	array<object>	Plain English explanations with tone ('good'|'warn'|'bad'|'info') and refIds
matches[]	array<object>	Match details: matchStrength (0-100), matchedPhrases, relation, reference
references[]	array<object>	Verified external citations conforming to Reference schema
checks[]	array<object>	Transparency list of all executed and skipped checks
disclaimer	string	Standard: 'Lie Detector Over Media gives signals, not final proof.'

4.2 Reference Verification Standards

- **FR-4.1 Schema Completeness:** Every reference must include id, publisher, title, rating, ratingMeaning, claim, url, date, verifiedOn, and verificationStatus.
- **FR-4.2 Stale Notice Warning:** References older than 2 years are flagged with 'This fact-check is old - check for updates'.
- **FR-4.3 Automated Verification:** Scripts (scripts/verifyReferences.js) must perform live HTTP GET checks confirming status 200 OK before deployment.

5. Non-Functional & Security Requirements

- **NFR-1 Security & Privacy:** Zero persistent storage. Memory buffers only. Helmet security headers enabled. IP rate limit: 30 requests per 15 minutes per IP.
- **NFR-2 Upload Restrictions:** Strict 4 MB client and server upload ceiling. HTTP 413 returned gracefully on oversized payloads.
- **NFR-3 Response Latency:** Local heuristic analysis must complete in < 2,500 ms. Server enforces a 25-second hard timeout abort.
- **NFR-4 Accessibility & UI:** Warm paper palette (#FFFDF7 cream, #8B5E3C brown, #F6C453 gold). High contrast ratios. Dedicated print stylesheet for PDF exports.
- **NFR-5 Code Quality & Testing:** TypeScript strict mode (0 any types). Backend Node test runner and frontend Vitest must maintain 100% pass rates.

Copyright (c) 2026 Sarvesh Pandey. All rights reserved. (MIT License)

6. Quality Assurance & Acceptance Criteria

Suite	Test Target	Verification Criteria & Passing Status
Backend API	Status & Routes	Validates /api/health, HTTP 400 empty/oversized inputs, CORS (Passed: 8/8)
Claim Matcher	False Positive Rate	35 benign claims evaluated: 0 false positives (0.00% <= 2.0% threshold)
Claim Matcher	Rumor Recognition	Verified specific rumors match with score >= 0.60 (Passed: 10/10)
Confidence	Level Rules	High/medium/low rule assertions verified (Passed: 5/5)
References	Dead Link Checker	Mock HTTP 200/404 unit tests + 8 live URLs verified 200 OK (Passed)
Frontend UI	Components & Flow	Vitest suites for Verdict, Reasons, MatchDetails, References (Passed: 26/26)
Code Standards	ESLint & TSC	Zero ESLint warnings, zero TypeScript errors, clean production bundle

DOCUMENT APPROVAL & SIGN-OFF

Product Name: Lie Detector Over Media ('See what's real.')

Author & Lead Engineer: Sarvesh Pandey

Copyright: Copyright (c) 2026 Sarvesh Pandey. All rights reserved.

Official GitHub Repository: https://github.com/xsarveshpandey21/lie-detector-over-media

Verification Status: Passed All 43 Backend & 26 Frontend Tests (100% CI Green)

Approved for deployment and release on October 30, 2026.

Copyright (c) 2026 Sarvesh Pandey. All rights reserved. (MIT License)